

macOS Application Firewall & Network Traffic Analysis Using Little Snitch

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Summary

This project evaluated outbound network activity using Little Snitch on macOS. To conduct the analysis, a baseline was implemented using applications such as Visual Studio and Google Chrome. The configuration of those applications were tested to determine whether network access could be restricted.

Objectives

1. Establish a baseline of normal outbound network activity
2. Identify applications communicating with external hosts
3. Analyze destination hosts and network ports
4. Configure application hosts and network ports
5. Configure application-level firewall rules
6. Test whether firewall rules successfully block traffic

Tools and Environment

Tool	Purpose
macOS	Endpoint under investigation
Little Snitch	Application-Level Firewall
Little Snitch Network Monitor	Network Traffic Monitoring
Google Chrome	Example application being analyzed
Visual Studio Code	Development application being analyzed
Terminal	Supporting network verification

Baseline Network Analysis

During the baseline monitoring period, multiple applications generate outbound network connections. Google Chrome generates connections to several external web services while browsing. Visual Studio Code also establishes connections to external services associated with development functionality.

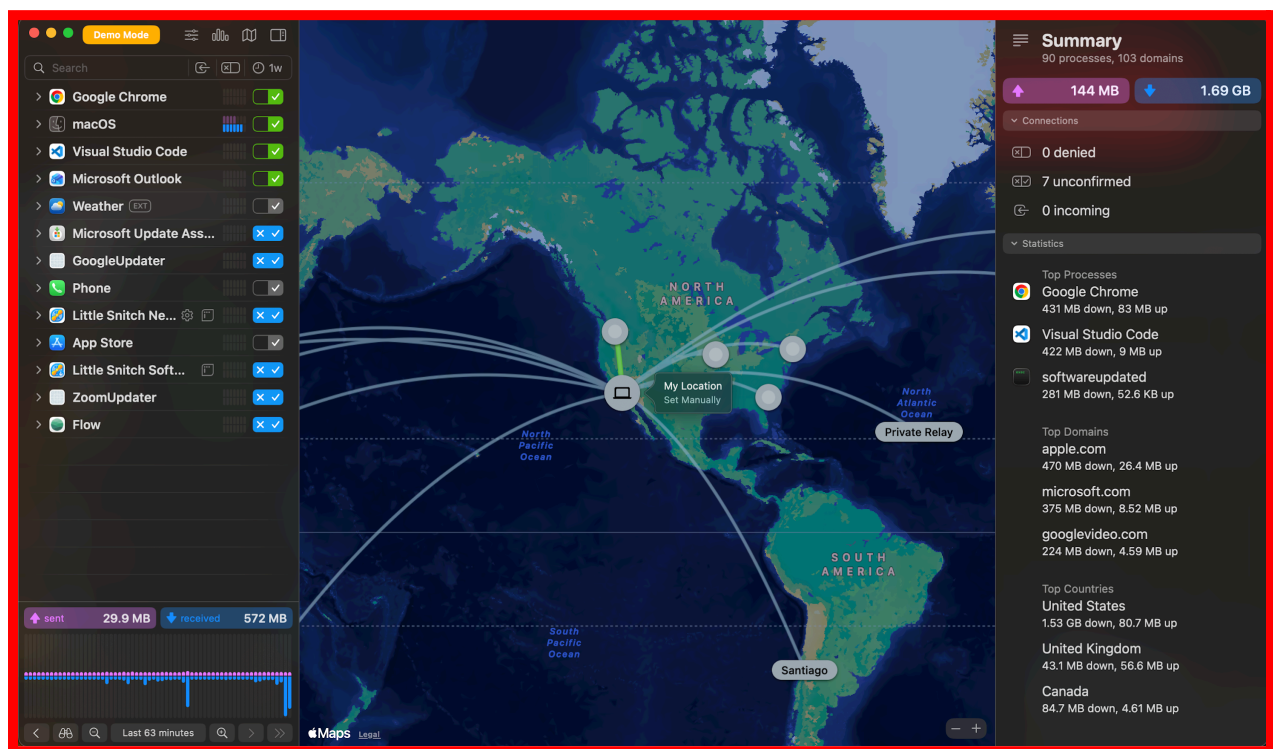


Figure 1 - Active Application Network Connections

Connection Analysis

Upon analysis, Chrome established multiple outbound connections during normal web browsing. Connections using TCP port 443 were consistent with encrypted HTTPS communications. The observed destinations were associated with normal browsing and were not considered suspicious based on the evidence.

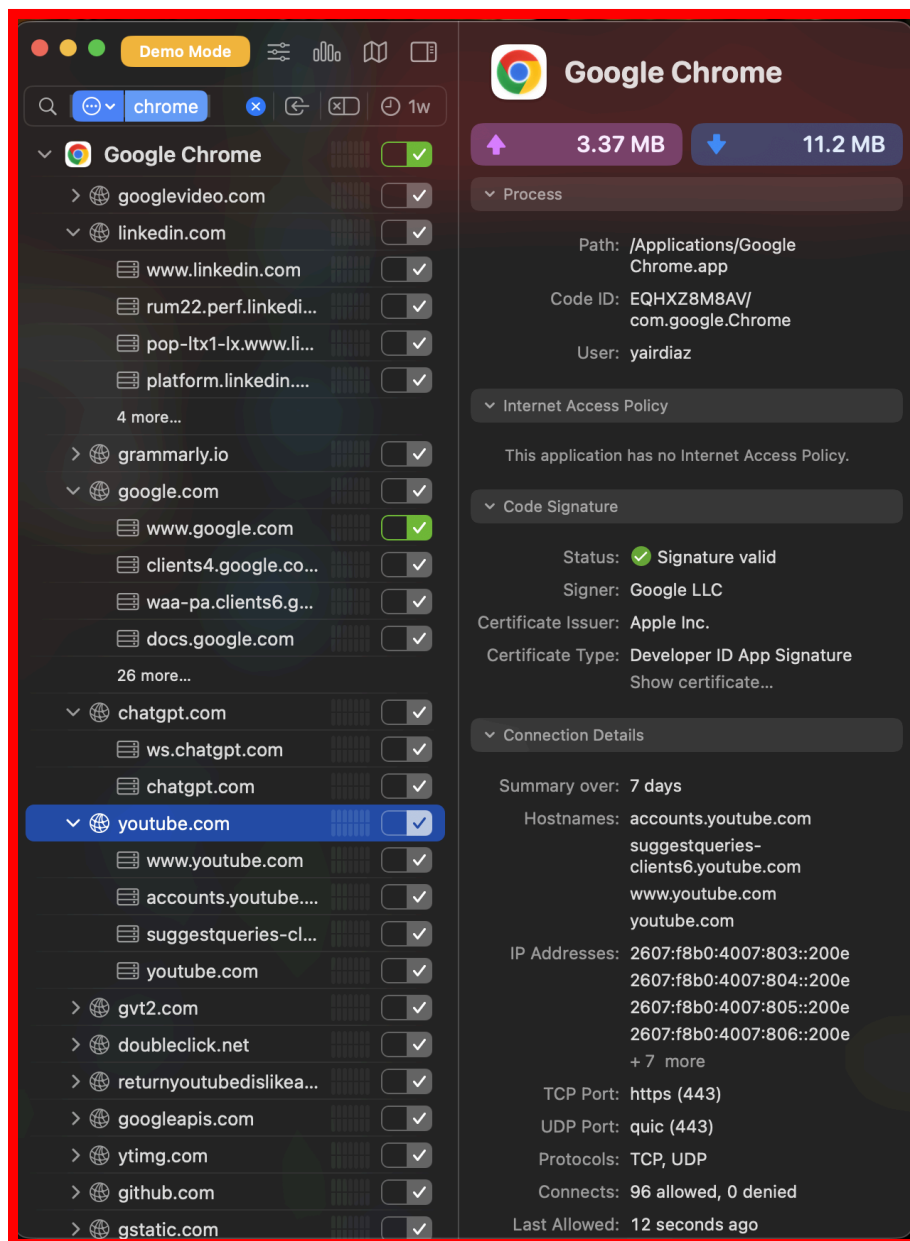


Figure 2 - Chrome Connection

Firewall Rule Configuration

Test Rule

Application: Google Chrome

Direction: Outbound

Action: **Deny**

Purpose: Determining whether Little Snitch could prevent the application from establishing external network connections.

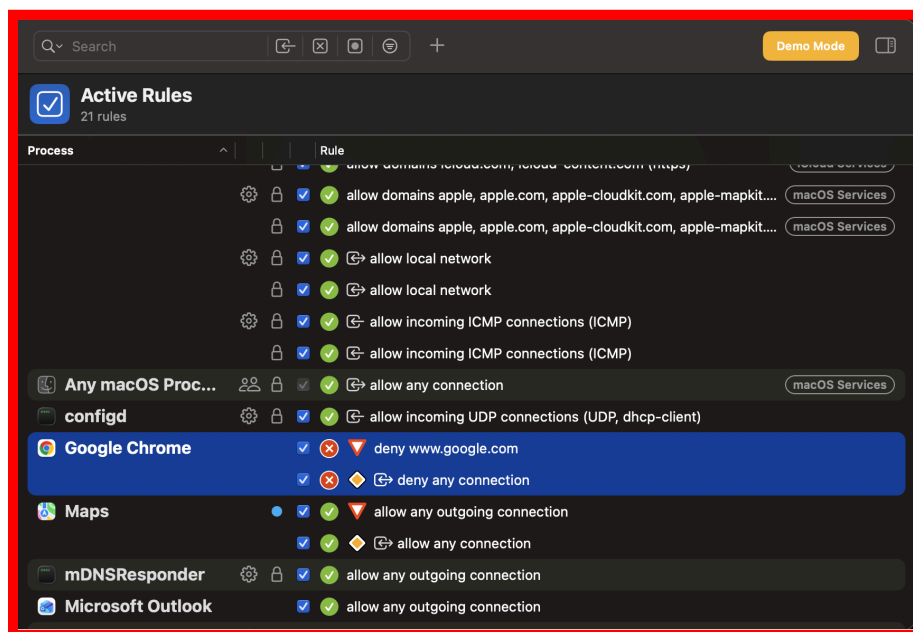


Figure 3- Firewall Rule

Firewall Testing & Results

Test Procedure

1. Created the deny rules
2. Opened the selected application
3. Attempted to generate network traffic
4. Monitored Little Snitch
5. Confirmed that the connection was denied

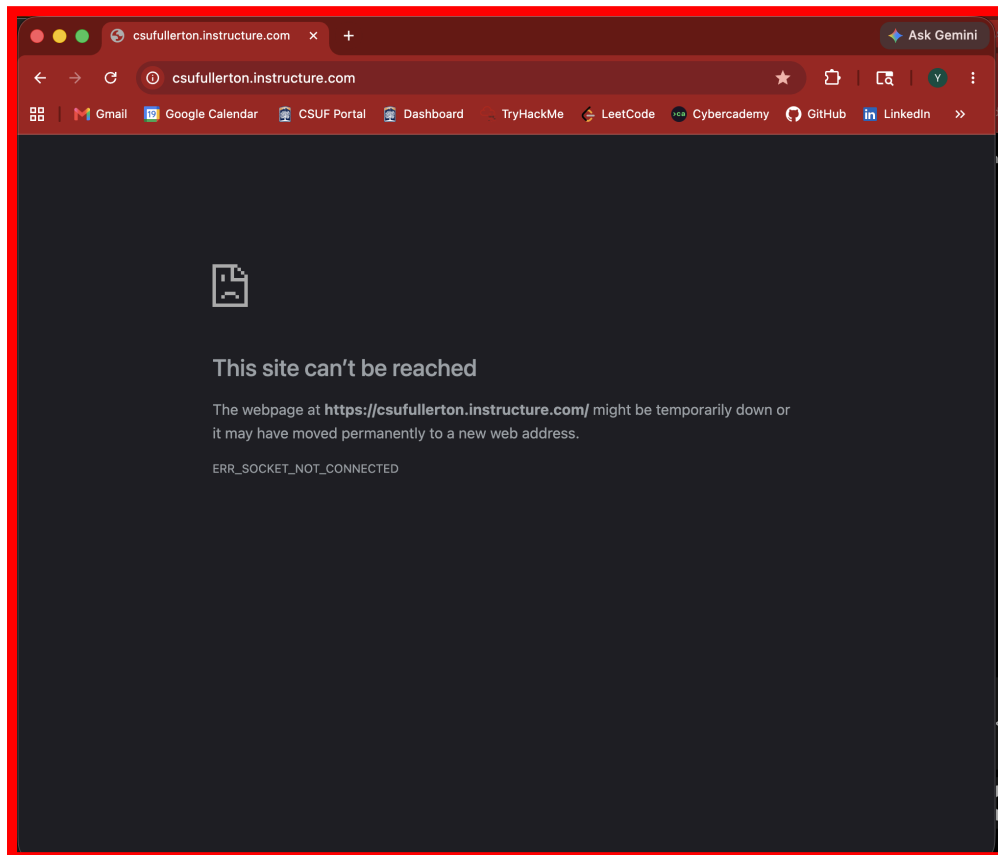


Figure 4 - Blocked Connection